

Closed-Form ZK: Efficient Temporal Proofs via CfC Neural Networks

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Abstract

We show that Closed-form Continuous-time (CfC) neural networks enable significantly more efficient zero-knowledge temporal proofs than ODE-based alternatives. The CfC closed-form solution $h(t) = (1 - \sigma(f)) \cdot h(0) + \sigma(f) \cdot x_\infty$ decomposes into exactly **1 AND constraint per neuron per timestep** in Binius (a binary-field STARK system), with all additive state updates free. With in-circuit sigmoid approximation, this increases to 2 AND per step. By contrast, Liquid Time-Constant (LTC) networks with RK4 integration require ~ 16 AND per step. We benchmark real Binius proofs at two scales: 8 neurons $\times$ 10 timesteps (96 AND, 493 ms) and 64 neurons $\times$ 100 timesteps (7,360 AND, 954 ms without sigmoid; 13,760 AND, 680 ms with sigmoid). These results demonstrate that CfC’s algebraic structure, combined with binary-field proof systems, makes verifiable temporal neural computation practical for the first time.

1 Introduction

Temporal neural networks model continuous-time dynamics in time-series data. Liquid Time-Constant (LTC) networks [?] and their closed-form counterpart CfC [?] have shown strong performance in autonomous driving, medical monitoring, and robotics. A growing need exists for *verifiable* temporal computation: proving that a neural network processed a time series correctly without revealing the input data.

Zero-knowledge proofs (ZKPs) can provide this guarantee, but the computational cost of proving ODE integration is prohibitive. Each RK4 substep requires multiple multiplications per neuron, and $S \approx 16$ substeps per timestep is typical for numerical accuracy. For N neurons over T timesteps, an ODE-based LTC requires $O(16 \cdot N \cdot T)$ non-linear constraints.

CfC networks eliminate this overhead with a closed-form solution that replaces iterative integration with a single expression per timestep. We show that this algebraic simplification has a direct correspondence in binary-field STARK proof systems (Binius [?]), where the CfC update decomposes into:

$$\delta = x_\infty \oplus h \quad (\text{XOR: free in Binius}) \quad (1)$$

$$p = \sigma \wedge \delta \quad (\text{AND: 1 constraint}) \quad (2)$$

$$h' = h \oplus p \quad (\text{XOR: free}) \quad (3)$$

2 Background

CfC Networks. The CfC solution [?] is:

$$h(t) = (1 - \sigma(f(t))) \cdot h(0) + \sigma(f(t)) \cdot x_\infty$$

where σ is sigmoid and $f(t) = -\Delta t/\tau$. This is an $O(1)$ expression per neuron per timestep — no numerical integration needed.

Binius. A STARK proof system over binary tower fields $\text{GF}(2^k)$ [?]. Addition (XOR) is native field addition and costs zero non-linear constraints. Only AND/MUL operations incur constraints.

3 CfC Temporal Proof Circuit

3.1 Without Sigmoid (Witness-Supplied)

When σ is provided as a precomputed witness value, the CfC update requires exactly **1 AND per neuron per timestep**: the multiplication $\sigma \wedge \delta$. Both XOR operations are free in Binius.

3.2 With In-Circuit Sigmoid

We implement sigmoid as a piecewise mask approximation:

$$\sigma(f) \approx f \wedge 0\mathbf{x}\mathbf{FF}$$

This extracts the low 8 bits of f as a crude sigmoid output, requiring 1 additional AND per step. Total: **2 AND per neuron per timestep**.

A production-grade piecewise linear sigmoid (5–10 segments) would require 5–10 AND per step, still far fewer than the ~ 16 for ODE-based LTC.

4 Experimental Results

All benchmarks use Binius64 [?] with real cryptographic proof generation and verification. Hardware: AMD Ryzen, 32 GB RAM, NixOS.

Table 1: CfC Temporal Proofs in Binius (Real Proofs)

Config	AND	XOR (free)	Prove	Size
<i>Without sigmoid (witness-supplied σ)</i>				
8N $\times$ 10T	96	160	493 ms	478 KB
64N $\times$ 100T	7,360	12,800	954 ms	156 KB
<i>With in-circuit sigmoid ($\sigma = f \wedge 0\mathbf{x}\mathbf{FF}$)</i>				
64N $\times$ 100T	13,760	12,800	680 ms	166 KB
<i>Estimated: LTC + RK4 (16 substeps per timestep)</i>				
64N $\times$ 100T	$\sim 117,760$	—	$\sim 10\text{s}+$	—

The 64-neuron, 100-timestep CfC proof generates in under 1 second with 7,360 AND constraints. An equivalent LTC+RK4 circuit would require $\sim 117,760$ AND constraints (16 $\times$ more).

5 Limitations

- Our sigmoid approximation ($f \wedge 0xFF$) is crude. A numerically accurate piecewise linear sigmoid would add 5–10 AND per step instead of 1.
- We benchmark temporal evolution in isolation. A full neural inference proof would also need to prove input encoding and output decoding.
- The LTC+RK4 comparison is estimated, not measured. Implementing a full ODE-based temporal proof circuit is future work.
- Binius64 is pre-1.0 software; API and performance may change.

6 Related Work

zkRNN [?] proves RNN execution but does not exploit closed-form temporal solutions. SUMMER [?] adds recursive ZKPs for RNN training. Neither addresses the CfC/LTC distinction or binary-field alignment. Our companion paper [?] establishes the broader algebraic alignment between HDC and binary-field STARKs.

7 Conclusion

CfC’s closed-form solution translates directly into efficient binary-field STARK circuits: 1–2 AND per neuron per timestep, with all additive operations free. This is $\sim 16\times$ fewer constraints than ODE-based temporal networks, making verifiable neural computation practical for temporal data processing.

References

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